

Complete SQL Notes (Beginner to Advanced)

1. Introduction to SQL

SQL (Structured Query Language) is used to create, retrieve, update and delete data stored in databases.

2. SQL Command Types

DDL: CREATE, ALTER, DROP, TRUNCATE

DML: INSERT, UPDATE, DELETE

DQL: SELECT

DCL: GRANT, REVOKE

TCL: COMMIT, ROLLBACK, SAVEPOINT

3. CREATE TABLE

```
CREATE TABLE Student(  
ID INT PRIMARY KEY,  
Name VARCHAR(50),  
Age INT  
);
```

4. INSERT

```
INSERT INTO Student VALUES(101,'Ravi',22);
```

5. SELECT and WHERE

```
SELECT * FROM Student;
```

```
SELECT * FROM Student WHERE Age > 20;
```

6. Operators

Comparison: =, >, <, >=, <=, <>

Logical: AND, OR, NOT

BETWEEN, IN, LIKE, IS NULL

7. Aggregate Functions

COUNT(), SUM(), AVG(), MAX(), MIN()

8. GROUP BY and HAVING

Used to group rows and filter grouped data.

9. UPDATE and DELETE

UPDATE Employee SET Salary=50000 WHERE ID=101;
DELETE FROM Employee WHERE ID=101;

10. Joins

INNER JOIN
LEFT JOIN
RIGHT JOIN
FULL JOIN
SELF JOIN
CROSS JOIN

11. Subqueries and Set Operators

UNION, UNION ALL, INTERSECT, EXCEPT/MINUS

12. Views, Indexes and Sequences

Improve reusability and performance.

13. Transactions

COMMIT, ROLLBACK, SAVEPOINT

14. Stored Procedures, Functions and Triggers

Used to automate database tasks.

15. CTE and Window Functions

ROW_NUMBER(), RANK(), DENSE_RANK()

16. Normalization

1NF, 2NF, 3NF and BCNF.

17. Important Interview Topics

Primary Key vs Foreign Key
DELETE vs TRUNCATE vs DROP
Joins and Subqueries
Group By vs Having
Views and Indexes